

1. Part 1: Sizing Chart

Design Specifications

Table 1: CS Amplifier Design Specifications

Specification	0.18 μm CMOS
DC Gain $ A_v $	8
Supply V_{DD}	1.8 V
Current Consumption I_D	60 μA
L	2 μm

Gain Expression and V^* Definition

The voltage gain of a resistive-loaded CS amplifier is:

$$|A_v| \approx g_m R_D = \frac{2I_D}{V_{ov}} \cdot R_D = \frac{2V_{RD}}{V_{ov}}$$

For a real MOSFET the square-law overestimates g_m . We define V^* using actual simulation data:

$$V^* = \frac{2I_D}{g_m} \Leftrightarrow g_m = \frac{2I_D}{V^*}$$

The corrected gain expression (used throughout this lab) is:

$$|A_v| \approx \frac{2V_{RD}}{V^*}$$

Determine R_D

To maximise output swing, the common-mode (DC) output level is set to $V_{DD}/2$, so $V_{RD} = V_{DD}/2$. Given $I_D = 100 \mu\text{A}$:

$$R_D = \frac{V_{RD}}{I_D} = \frac{V_{DD}/2}{I_D}$$

Derivation of R_D

$$\text{0.18 } \mu\text{m CMOS } (V_{DD} = 1.8 \text{ V}): R_D = \frac{0.9 \text{ V}}{60 \mu\text{A}} = 15 \text{ k}\Omega$$

Calculate Required V_Q^*

Derivation of V_Q^* for both technology nodes

0.18 μm CMOS: $V_Q^* = \frac{1.8}{8} = 0.225 \text{ V}$

V^* and V_{ov} vs V_{GS} (Overlaid Plot)

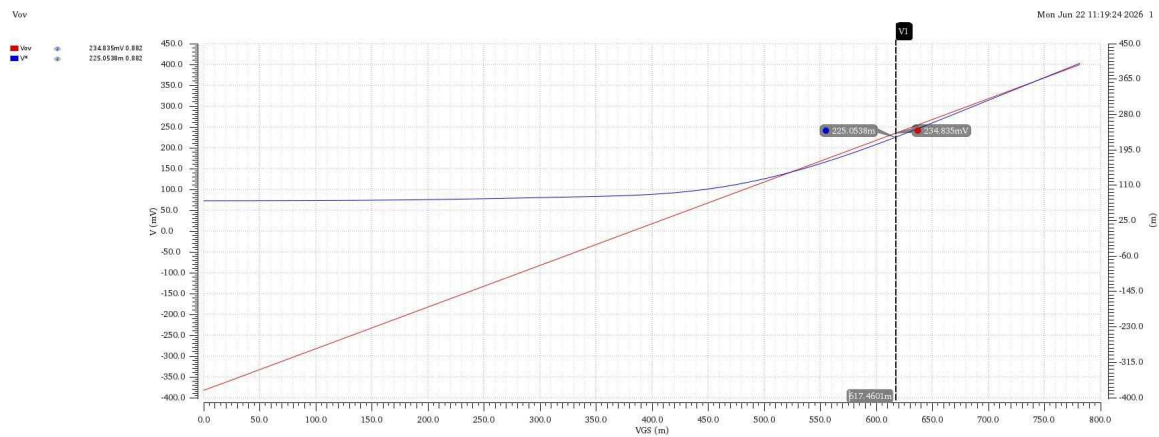


Figure 1: Observations on V^* vs V_{ov}

Locate V_Q^* on the Sizing Chart

From the chart above, locate the V_{GS} point where $V^* = V_Q^*$ and read off V_{ovQ} and V_{GSQ} .

Table 2: Operating-Point Voltages from Sizing Chart

Parameter	0.18 μm
$V_Q^* (mV)$	225
$V_{ovQ} (mV)$	234.835
$V_{GSQ} (mV)$	617.4601

$I_D, g_m, g_{ds} \text{ vs } V_{GS}$

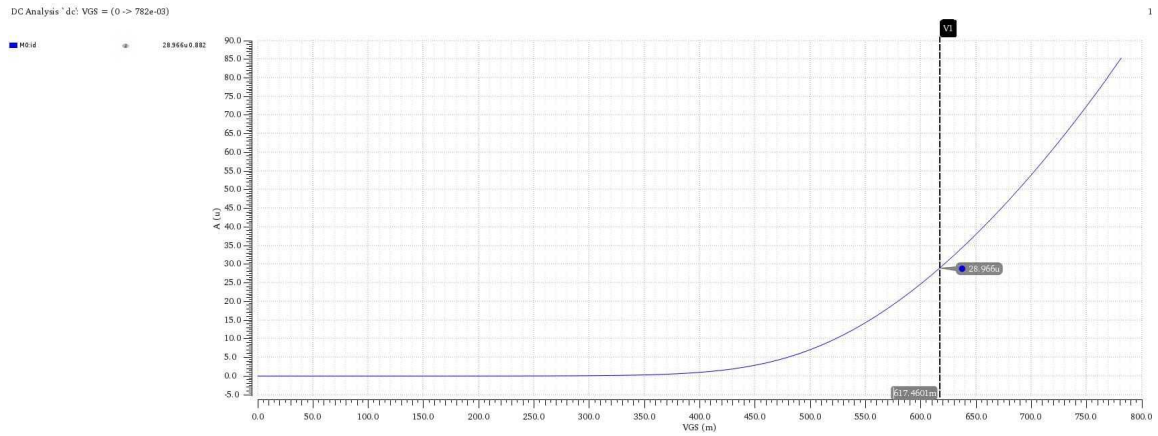
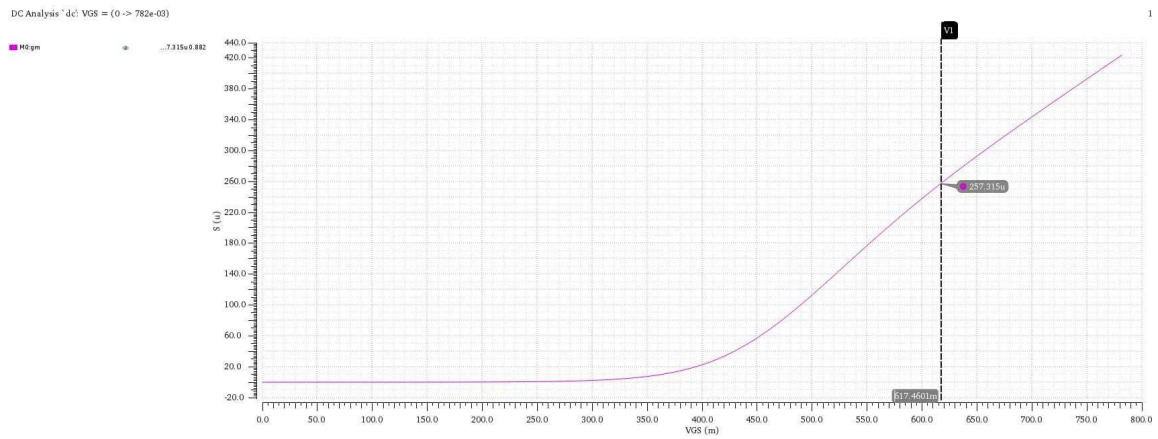
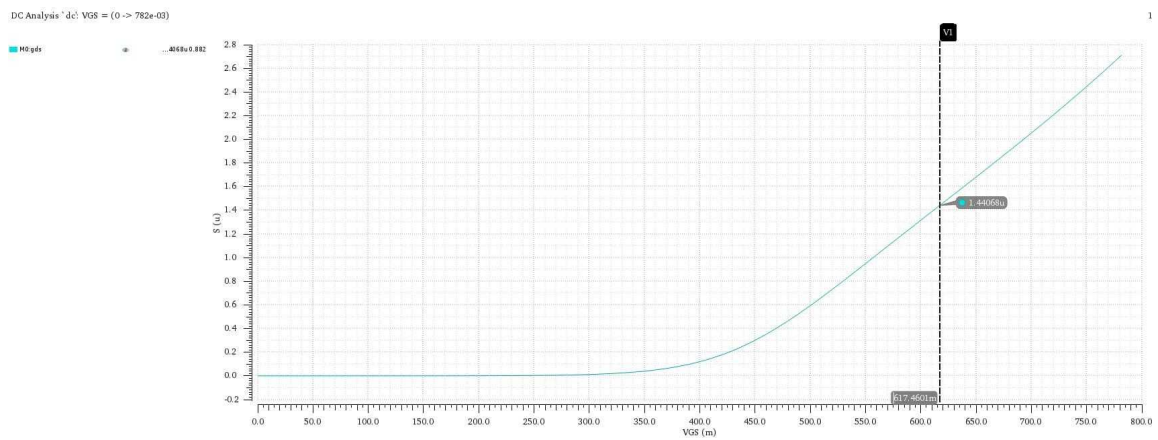
Figure 2: I_D vs V_{GS} Figure 3: g_m vs V_{GS} Figure 4: g_{ds} vs V_{GS}

Table 3: Device Parameters at V_{GSQ} for Reference Width $W = 10\ \mu\text{m}$

Parameter	0.18 μm
I_{DX} at V_{GSQ}	28.966 μA
g_{mX} at V_{GSQ}	257.315 $\mu\text{A}/\text{V}$
g_{dsX} at V_{GSQ}	1.44068 μS

Scale Width W to Achieve $I_{DQ} = 60\ \mu\text{A}$

Since $I_D \propto W$ at constant V_{GS} and L , we cross-multiply:

$$\frac{W}{10\ \mu\text{m}} = \frac{I_{DQ}}{I_{DX}} \Rightarrow W = \frac{I_{DQ}}{I_{DX}} \times 10\ \mu\text{m}$$

Table 4: Width Scaling (Cross-Multiplication Table)

W	I_D
10 μm	28.966 μA
$W = 20.7139\ \mu\text{m}$	60 μA

Scale g_{mQ} and g_{dsQ} ; Verify Gain

$g_m \propto W$ and $r_o = 1/g_{ds} \propto 1/W$ at constant V_{ov} and L , so:

$$g_{mQ} = g_{mX} \times \frac{W}{10\ \mu\text{m}}, \quad g_{dsQ} = g_{dsX} \times \frac{W}{10\ \mu\text{m}}$$

Gain verification:

$$A_v = -g_{mQ}(R_D \parallel r_o), \quad r_o = \frac{1}{g_{dsQ}}$$

Calculation of g_{mQ} , g_{dsQ} , r_o , and A_v

0.18 μm CMOS:

$$g_{mQ} = g_{mX} \times \frac{W}{10\ \mu\text{m}} = 257.315\ \mu\text{A}/\text{V} \times \frac{20.7139}{10} = 532.999\ \mu\text{A}/\text{V}$$

$$g_{dsQ} = g_{dsX} \times \frac{W}{10\ \mu\text{m}} = 1.44068\ \mu\text{S} \times \frac{20.7139}{10} = 2.984\ \mu\text{S}$$

$$r_o = \frac{1}{g_{dsQ}} = 335.098\ \text{k}\Omega$$

$$|A_v| = -532.999\ \mu\text{S} * (15\ \text{k}\Omega \parallel 335.098\ \text{k}\Omega) = 7.6524\ \text{V}/\text{V}$$

Does the calculated gain meet the specification? Comment on the agreement.

Yes, There is margin of 4.3445% in the gain.

2. Part 2: CS Amplifier

2.1 OP and AC Analysis

CS Amplifier Testbench

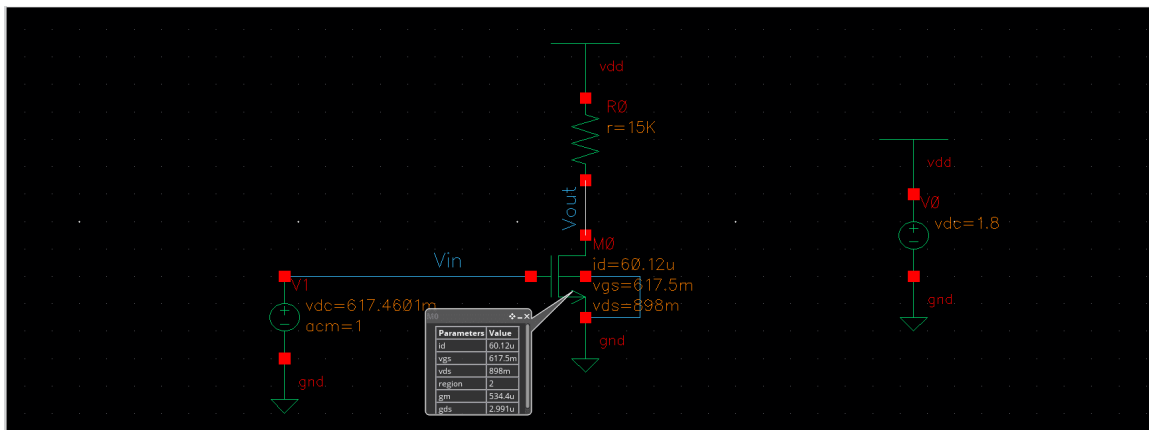


Figure 5: CS Amplifier Schematic

DC Operating Point Simulation

Table 5: DC Operating Point — Simulation vs Part 1 Design

Parameter	Unit	Simulation	Part 1 Design
I_D	μA	60.12	60
V_{GS}	mV	617.5	617.4601
V_{DS}	mV	898	900
g_m	$\mu\text{A}/\text{V}$	534.4	532.999
g_{ds}	μS	2.991	2.984
r_o	$\text{k}\Omega$	334.336	335.098
Region	—	2(sat)	2 (sat)

Yes, they are very close.

Compare r_o and R_D

Is ignoring r_o justified? What changes at minimum L ?

Yes, It is justified as

$$|gmR_D| = 7.995V/V$$

If we used minimum L , r_o will decrease, So the gain will deteriorate below 8 and $R_D \parallel r_o$ can't be ignored. Error in gain due to ignoring r_o will increase.

Intrinsic Gain

Calculate intrinsic gain $A_0 = g_m \cdot r_o$

$$|A_0| = g_{mQ} \times r_o = 532.999 \mu A/V \times 335.098 k\Omega = 178.61 \text{ V/V}$$

Amplifier Gain (Analytical)

Calculate $A_v = -g_m(R_D \parallel r_o)$ and compare with A_0

$$R_D \parallel r_o = \frac{R_D \cdot r_o}{R_D + r_o} = 14.357 k\Omega$$

$$A_v = -g_{mQ} \cdot (R_D \parallel r_o) = -7.652$$

Relation between $|A_v|$ and A_0 : $|A_v| \ll A_0$

AC Analysis (1 Hz – 1 GHz)

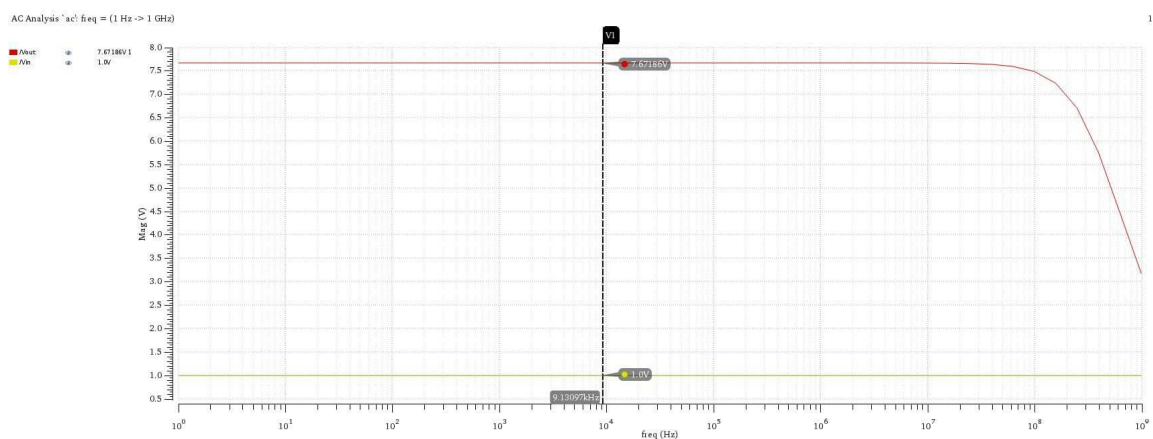


Figure 6: Small signal gain vs frequency

2.2 Gain Non-Linearity

V_{OUT} vs V_{IN} (DC Sweep, $0 \rightarrow V_{DD}$, step 2 mV)

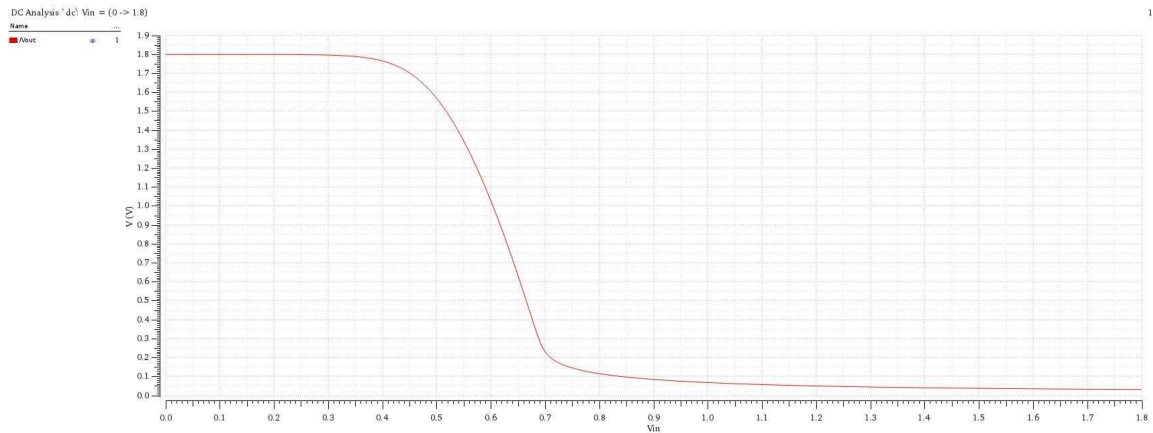


Figure 7: V_{out} vs V_{in}

Is V_{out} vs V_{in} linear? Why?

No, V_{OUT} vs V_{IN} is not linear. For the amplifier to be perfectly linear, the voltage transfer characteristic would need to be a straight line with a constant slope, which would indicate a constant voltage gain. Instead, the graph shows a distinct curve. Physically, this occurs because the MOSFET's transconductance (g_m) depends heavily on the input voltage. As V_{IN} sweeps from 0 to 1.8 V, the transistor transitions through cutoff, saturation, and triode regions, causing the gain to vary continuously and resulting in a non-linear transfer curve.

Derivative of V_{OUT} (Small-Signal Gain) vs V_{IN}

The derivative $\frac{dV_{OUT}}{dV_{IN}}$ is the local small-signal gain.

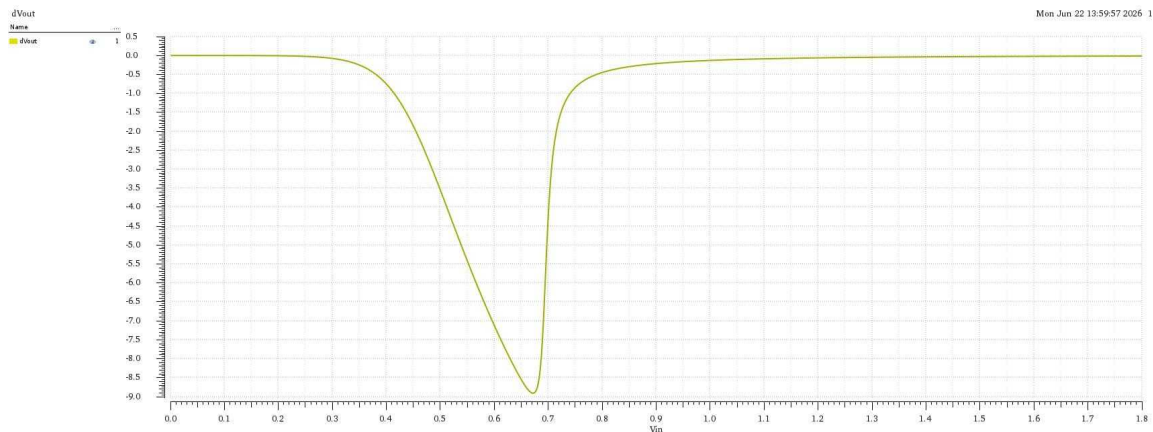


Figure 8: dV_{OUT}/dV_{IN} vs V_{IN}

Is the gain constant with V_{IN} ? What does gain variation imply?

No the gain is not constant with V_{in} , Gain variation implies that the CS single stage amplifier is nonlinear.

Transient Simulation: g_m vs Time

Input stimulus: $v_{in}(t) = V_{GSQ} + 10 \text{ mV} \cdot \sin(2\pi \cdot 1 \text{ kHz} \cdot t)$. Run transient simulation for 2 ms.

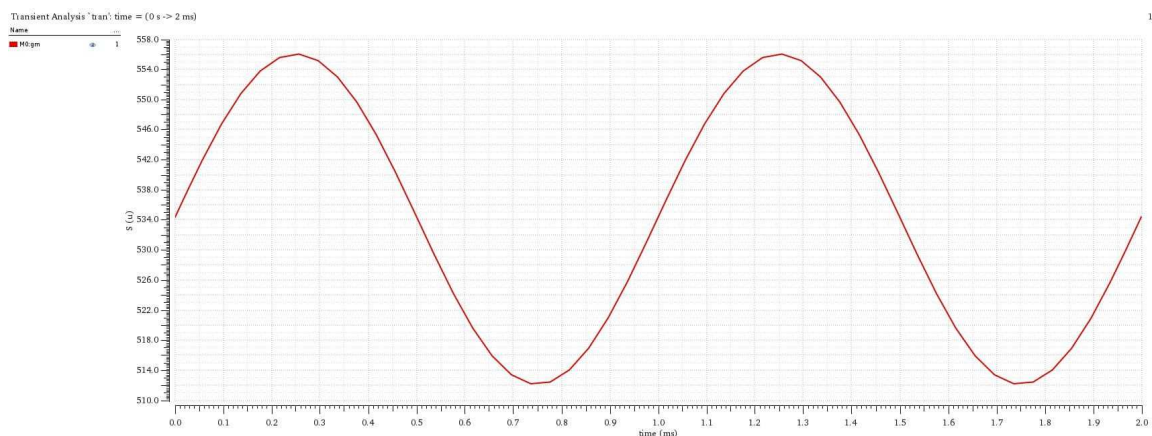


Figure 9: g_m vs Time

Does g_m vary with the input signal? What does this mean for linearity?

Yes, g_m varies. a varying g_m means the gain is constantly changing as the input signal swings. This dynamic changing of gain distorts the output signal (introducing harmonics), proving that the large-signal behavior of the transistor is non-linear

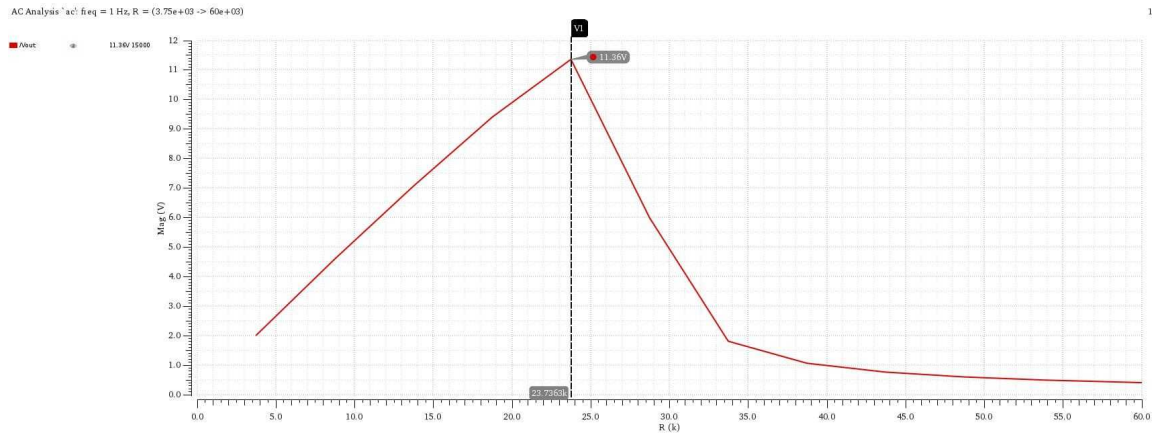
Linearity Conclusion

Is this amplifier linear? Justify with reference to g_m variation.

No, it is non-linear.

2.3 [Optional] Maximum Gain

AC analysis with a unit AC source; sweep design variable R_D from $R_D/4$ to $4R_D$ at $f = 1 \text{ Hz}$.

Figure 10: $|A_v|$ vs R_D

Explain why gain first increases then decreases with R_D .

Why the gain increases: Initially, the gain is directly proportional to the load resistor ($|A_v| \approx g_m R_D$). As long as $V_{DS} \geq V^*$, the MOSFET remains in the saturation region, and increasing R_D steadily increases the voltage gain.

Why the gain decreases: As R_D continues to increase, the DC voltage drop across it ($I_D R_D$) becomes so large that V_{DS} drops below V^* . This forces the MOSFET out of saturation and into the triode region. In triode, the intrinsic output resistance (r_o) completely collapses and transconductance (g_m) degrades. Since the total gain relies on the parallel combination ($R_D \parallel r_o$), the collapsed r_o dominates, causing the amplifier's gain to plummet.

Table 6: Maximum Gain Results

Quantity	Unit	Simulation	Analytical
Optimum R_D for max gain	k Ω	23.74	26.25
Maximum $ A_v $	V/V	11.36	12.97

At maximum gain the available output swing is

$$V_{DD} - V^* = 1.8 - 0.225 = 1.575V$$

Analytical derivation of optimum R_D and maximum $|A_v|$

$$R_D = \frac{1.575}{60\mu A} = 26.25k\Omega$$

$$|A_v| = g_m R_D = 12.97V/V$$

Is scaling down V_{DD} beneficial for gain? Comment.

No, scaling down the supply voltage is bad for gain. Although decreasing current might lower V^* , the voltage gain heavily depends on the voltage drop across the load resistor ($|A_v| \approx \frac{2V_{RD}}{V^*}$). Because V_{RD} is strictly limited by the supply voltage to keep the transistor in saturation ($V_{RD} \leq V_{DD} - V^*$), scaling down V_{DD} forces a smaller V_{RD} , significantly degrading the maximum achievable gain and drastically reducing the available output signal swing.

2.4 [Optional] Gain Linearisation (Feedback)

PMOS Active Load Sizing

Repeating NMOS testbench for the PMOS. $W_P = 82.56\mu m$

$$V_{SG} = 616.7mV$$

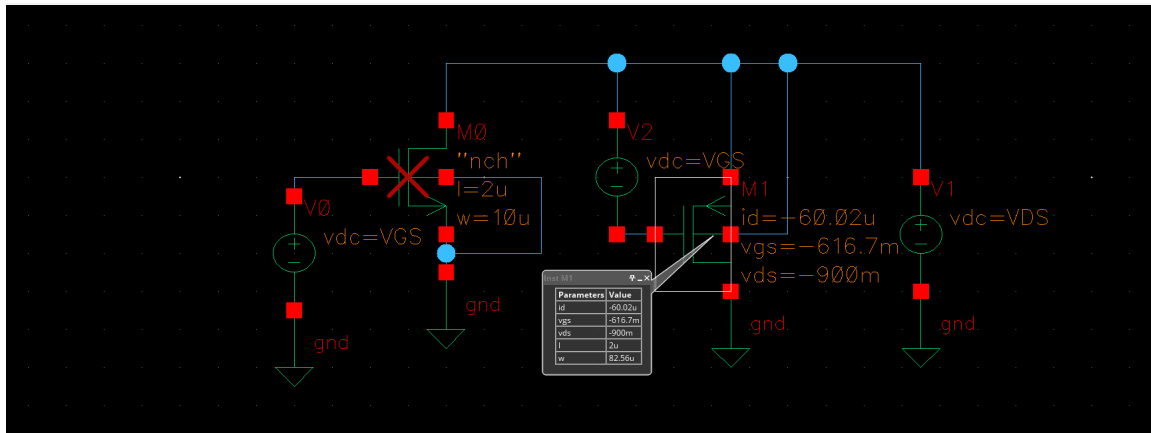


Figure 11: PMOS testbench

Schematic with PMOS Load and Feedback Resistors

$$R_f = 8M\Omega$$

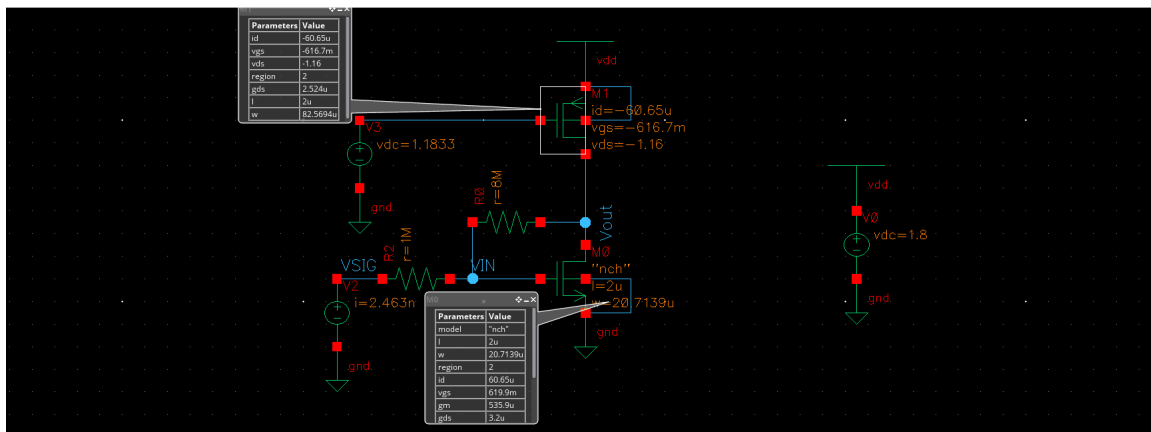


Figure 12: CS Amplifier with PMOS Active Load and Feedback Resistors

DC Sweep: V_{IN} and V_{OUT} vs V_{SIG}

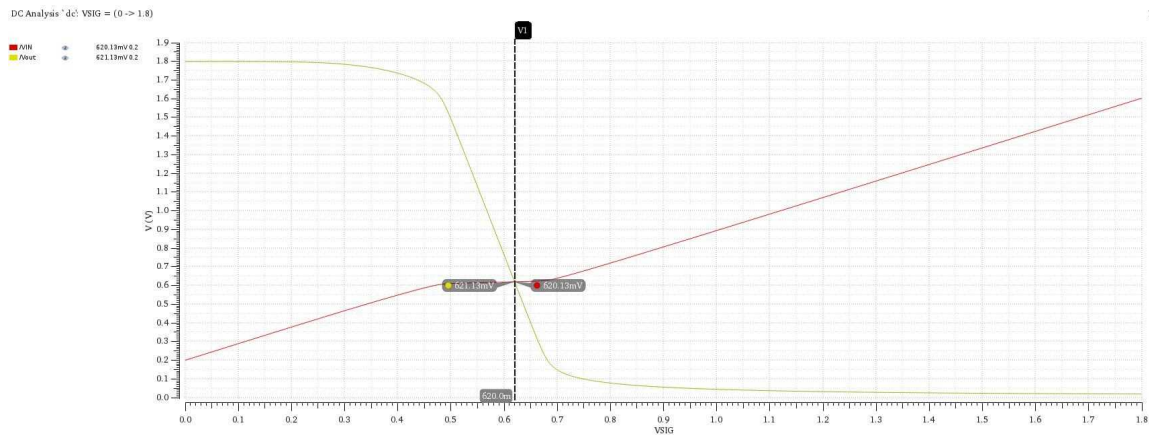


Figure 13: V_{IN} and V_{OUT} vs V_{SIG}

They cross at $V_{SIG} = 620mV$ similar to V_{GS} of M0 of $617.4601mV$ so the bias point stays constant as no current flows in R_{sig} and R_f , That is where the gain is linear.

Linearity of V_{OUT} vs V_{SIG}

Yes, it is more linear at the expense of less closed loop gain. This is because the negative feedback makes the overall gain depend mostly on the fixed resistors (R_f and R_{in}) rather than the transistor's internal transconductance (g_m). Since standard resistors are highly linear and do not change with voltage swings, the output becomes much more stable and free from distortion.

Derivative dV_{OUT}/dV_{SIG}

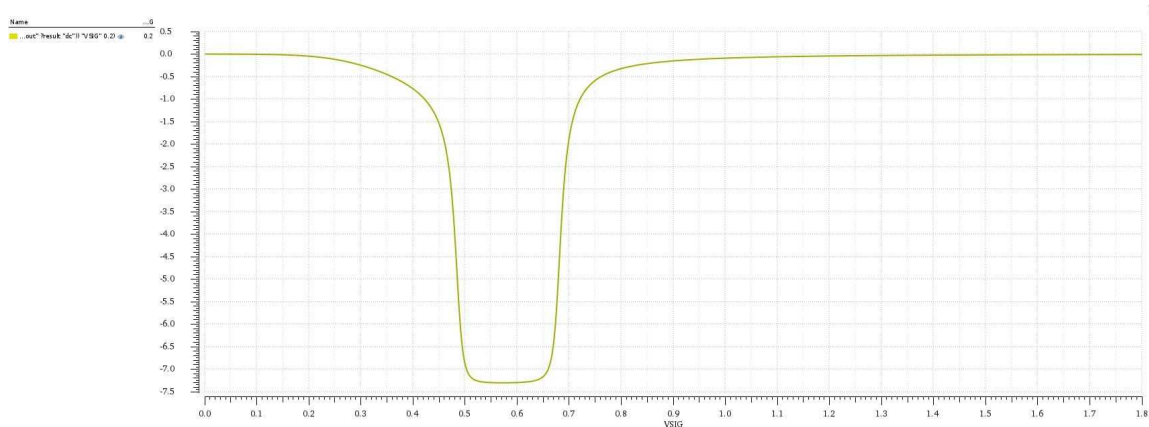


Figure 14: dV_{OUT}/dV_{SIG} vs V_{SIG}

Is the gain constant with V_{in} ?

Yes, the gain is significantly more constant, as evidenced by the wide, flat region in the derivative plot. Negative feedback desensitizes the circuit from the transistor's inherent non-linear transconductance (g_m). Instead, the overall closed-loop gain is dictated entirely by the passive feedback resistors. Because passive resistors maintain a constant value regardless of the applied voltage, the amplifier achieves a highly stable and linear transfer characteristic as long as the transistors remain in saturation.

Analytical Linear Input Range

The amplifier leaves saturation when $V_{DS} < V^*$. The output linear range is $\approx V_{DD} - 2V^*$. Dividing by gain:

$$\Delta V_{in} \approx \frac{V_{DD} - 2V^*}{|A_v|}$$

Calculate ΔV_{in} and compare with simulation

$$\Delta V_{in} \approx \frac{1.8 - 0.45}{8} = 0.16875V$$

From simulation $\Delta V_{in} = 0.16$

Table 7: Feedback Linearisation Summary

Parameter	Unit	Simulation	Analytical
Linear input range ΔV_{in}	mV	0.16	0.16875